

ISLAMIC UNIVERSITY OF TECHNOLOGY

BoardBazar, Gazipur, Dhaka, 1704, Bangladesh

SWE 4304- Software Project Lab

Final Project Report

Department of Computer Science And Engineering

BSc. in Software Engineering

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Title of the project:

SmartCDS

List of team members:

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Problem Statement:

The existing billing process in our university's canteen lacks efficiency and accuracy. There is a need for a comprehensive system that can streamline the billing process, provide accurate transaction records, and offer features for administrators, customers and salespersons to enhance their experience.

Potential Solution:

The SMARTCDS project aims to address the challenges in the current canteen billing system. By developing a user-friendly console application in C++, we aim to improve accuracy, track sales, and manage credit sales efficiently.

Implemented Features:

For Administrators:

1. User Authentication :

Successfully implemented a secure user authentication

system allowing users to log in and sign up based on their roles (Admin).

2. Food Management:

Add Food-

Developed a feature enabling administrators to add new food items to the canteen menu.

Delete Food -

Implemented the functionality to delete food items from the menu, providing administrators with control over the available options.

Edit Food -

Created a feature allowing administrators to modify existing food items, updating details such as name, price, or quantity.

Inventory Display-

Implemented a mechanism to display the current inventory, providing administrators with a comprehensive view of available food items.

For Salesperson:

1. User Authentication :

Successfully implemented a secure user authentication system allowing users to log in and sign up based on their roles (Salesperson).

2.Order Placement:

Buyers can place orders by selecting products, specifying quantities, and calculating the total bill.

3.Invoice Generation:

Generate itemized invoice with details such as product name, quantity, etc.

For Customers:

1. User Authentication :

Login-

Implemented a secure login mechanism for customers, ensuring authentication based on valid credentials.

Signup –

Developed a user-friendly signup process enabling customers to create accounts, providing essential information for identification.

2.Customer Profile Management:

Update Profile -

Implemented a feature allowing customers to update their profile information, such as username and mobile number.

Change Password -

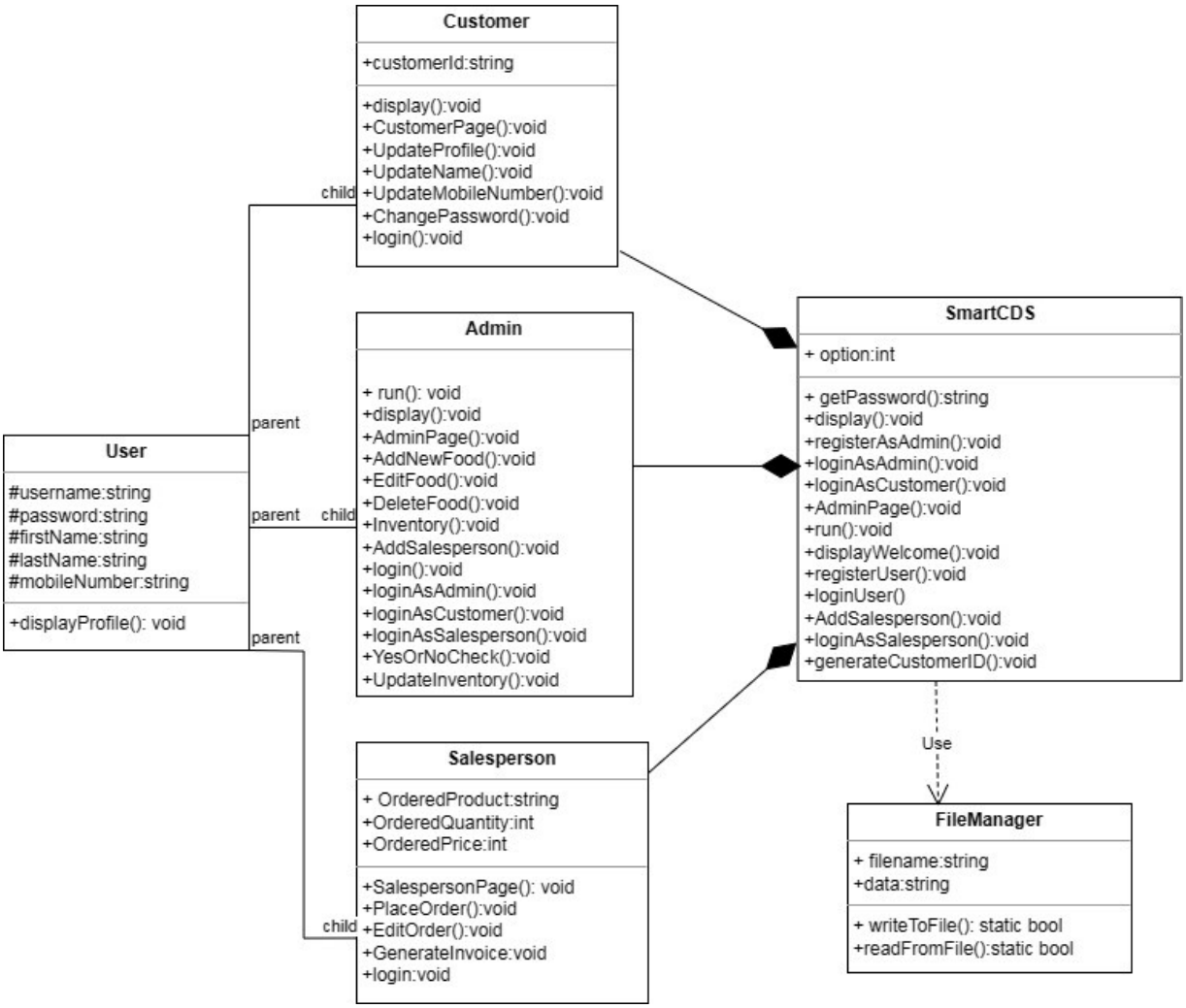
Provided customers with the ability to change their account password, enhancing security and user control.

3. Purchase History:

View Purchase History -

Implemented a functionality enabling customers to view their purchase history, displaying details of past transactions:

UML Diagram:



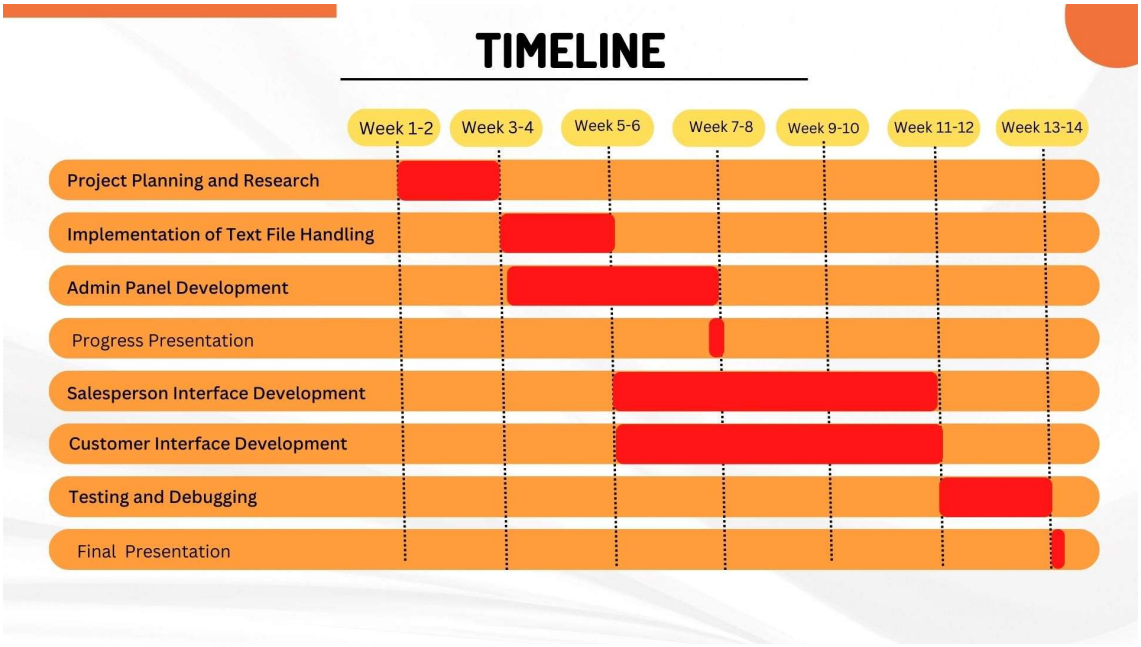
Tools and Technology:

- C++ for console application development.
- GitHub for version control and collaborative development.

Application Domain:

The application targets the university canteen billing system, aiming to enhance the efficiency and accuracy of transactions. It falls within the domain of point-of-sale systems and financial management for canteen operations.

Project Timeline:



Contribution:

We have distributed the tasks of this project in the following way:

Antara Arifa Mullick (ID:210042132):

Class: SmartCDS

(Full class implementation)

Class: Admin

(Full class implementation)

Niaz Rahman (ID:210042159):

Class: User

(Full class implementation)

Class: Salesperson (Full class implementation)

Labibah Hoque (ID:210042121):

Class: File Manager

(Full class implementation)

Class: Customer

(Full class implementation)

Future work:

The SMARTCDS project has potential extensions, including:

- Integration with online payment systems.
- Enhanced reporting features.
- Mobile application development for user convenience.

GitHub link:

https://github.com/niazbuoy08/SPL_1.git